Aayush Rai

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Education

Northeastern University Expected May 2028
Master of Science in Computer Science *Boston, MA*
Kansas State University August 2026
Bachelor of Science in Computer Science and Math Dual Major | GPA: 3.6 *Manhattan, KS*

- **Relevant Coursework:** Web Development, Data Structures and Algorithms, Object Oriented Programming, Probability & Statistics, Database Systems, Operating Systems, Computer Architecture, Software Design.

Skills

- **Languages:** Python, Java, C#, SQL, JavaScript, TypeScript, HTML/CSS
- **Frameworks:** React.js, Next.js, Node.js, FastAPI, ASP.NET Core, SQLAlchemy, REST APIs
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, ChromaDB, Pandas, NumPy
- **Tools & Databases:** Git, Docker, Linux, PostgreSQL, SQL Server, Vitest

Experience

Website Team Developer | *Hack for LA* July 2026 – Present

- Merged 4 PRs to hackforla.org across good-first, small, and medium tiers, including an audit of profile data for 30+ civic-tech teams that corrected leadership rosters and tech stacks, resulting in accurate listings for the site's 1,000+ monthly visitors.
- Reviewed 11 contributor PRs for YAML, formatting, and branching errors so that the merge team could approve changes faster and keep the review queue moving.
- Converted 3 epics and enhancement requests into scoped GitHub issues so that new volunteers had well-defined, ready-to-claim work in the prioritized backlog.

Machine Learning Researcher | *Kansas NSF EPSCoR (ARISE)* Nov 2025 – July 2026

- Engineered a RAG pipeline using LangChain, ChromaDB, and transformer-based embeddings to semantically index and query large-scale emergency response documents, achieving 90%+ retrieval precision.
- Architected a Gradio + FastAPI interface with asynchronous Python orchestration to handle concurrent stakeholder queries, keeping response latency under 10 seconds.

Computer Vision Researcher | *Kansas State University (CNAP)* Feb 2025 – May 2025

- Developed and trained pose estimation pipelines using YOLOv11, TensorFlow, and DeepLabCut on high-volume datasets, evaluating model accuracy with mean Average Precision across validation sets.
- Automated end-to-end data preprocessing with Python, OpenCV, Pandas, and NumPy so that raw experimental data could be standardized, resulting in 40% faster dataset labeling.

Projects

HeatSwitch | *Next.js, TypeScript, Claude API, Vitest* | Live Fall 2026

- Built and deployed a full-stack Next.js app solo at Hackachusetts 2026 that reads gas, electric, and oil bills with Claude vision so that Boston residents get a clear heat pump verdict with yearly savings, Mass Save rebates, and payback.
- Kept the LLM out of the math by validating extracted bill fields with Zod, then computing every dollar figure in pure TypeScript covered by 140+ unit and property-based tests, which means no number on screen is AI-generated.
- Cleaned and ranked 1,249 Boston multifamily buildings from the city's BERDO 2021 energy dataset by heat pump opportunity so that landlords and city staff could filter retrofit candidates by zip code.

BookFlix | *Full Stack Digital Library System* Fall 2025

- Architected a normalized SQL Server schema with indexing and ACID-compliant transactions to manage 2,000+ records, ensuring data integrity for concurrent user reviews and authentication.
- Developed backend services in ASP.NET Core (C#) using direct SQL execution for reproducible query behavior, and integrated a React + TypeScript frontend enabling search, reviews, authentication, and book discovery.

AguaCrop | *Kansas State TAPS Hackathon 1st Place* Fall 2024

- Built a RAG-based chatbot in Python with a Node.js backend API so that users could ask irrigation questions in plain language and get real-time, data-backed answers.
- Trained predictive models with Scikit-learn on historical agricultural and weather data and surfaced the output in a React.js frontend via REST APIs to visualize irrigation schedules, delivering a working MVP within a 48-hour sprint.